AI that computes with time and is already winning

A new family of learning machines, built for a world that happens as events, beats the best published models on public benchmarks with a fraction of their size.

5 of 5
datasets won on the public EasyTPP event-prediction leaderboard

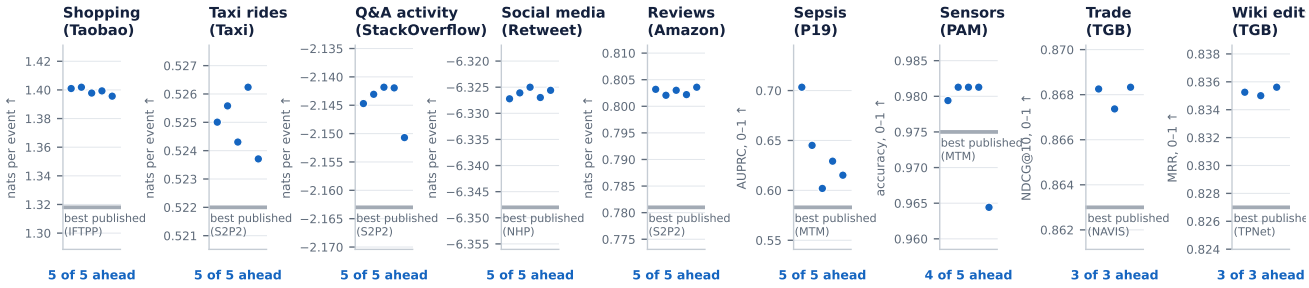
0.639
sepsis early-warning AUPRC vs 0.583 for the best published model ↑

15×
smaller and cheaper than the state-of-the-art model on Retweet, while more accurate

40 / 41
independent runs ahead of the best published result on the nine benchmarks won

In one sentence. Sleeping Machines is building a general-purpose computing substrate for machine intelligence, one that learns through time and sparse events, and its first public results are wins against the strongest published models, achieved by small, efficient models under strict, pre-registered test protocols.

40 of 41 independent runs ahead of the best published result



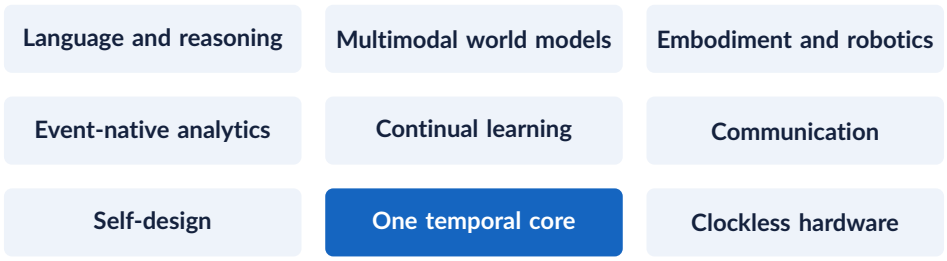
The benchmark wins use completed runs on official data, scored once on sealed test sets. The forward-credit development page reports separate private TRAIN-tail experiments, with its scope stated beside the numbers. Each benchmark's unit, and whether higher or lower is better, is stated beside every comparison (↑ higher is better, ↓ lower is better). Sources: `experiments/B1_EASYTPP.md`, `experiments/B2_IRREGULAR_TS.md`, `report/I_SCIENCE.md`.

THE AMBITION

One substrate for machine intelligence

Today's AI runs on a clock: every input is chopped into fixed steps, and the whole model works at every step whether or not anything happened. The real world is not like that. Purchases, heartbeats, sensor readings, messages, machine logs and words arrive as *events*, irregularly, with meaning in their timing and in their silences.

Sleeping Machines builds learning machines in which **time itself does part of the computation**. Our goal is one trainable substrate that spans:



What makes it different, in plain terms:

- **Memories that live through silence.** Each memory keeps evolving between events, so "how long since" and "what happened in between" are information, not missing data.
- **A race of clocks decides what happens next.** Possible next events race against each other; the one that fires first wins. This gives exact predictions of both *when* and *what*, and the same mathematics reproduces the attention mechanism inside today's large language models.
- **Only the relevant parts wake up.** Updates are sparse and addressed: the machine can hold far more capacity than it uses for any single event, which is where the name comes from.
- **Learning from roads not taken.** The model is also credited for the alternatives that lost the race, so it learns which routes to take.

> **Why this matters commercially.** Most of the world's valuable data is timestamped events: transactions, clinical records, telemetry, security and IT logs, user behaviour. Models that understand time natively can be more accurate on that data and much cheaper to run, on CPUs and edge devices as well as in datacentres. The same principles point to hardware without a global clock, where idle capacity costs almost nothing.

THE PROVING GROUND

The benchmarks, in plain language

We compete on established public benchmarks, using their official data splits and scoring rules, against the best results other teams have published. The headline event benchmark is **EasyTPP** (ICLR 2024), the standard test suite for predicting *when* the next event happens and *what* it will be. Its current leader is **S2P2** (NeurIPS 2025, GE HealthCare and UC Irvine), a modern state-space model. The **Temporal Graph Benchmark** (TGB, NeurIPS 2023) tests who interacts with whom, and when, against temporal graph neural networks.

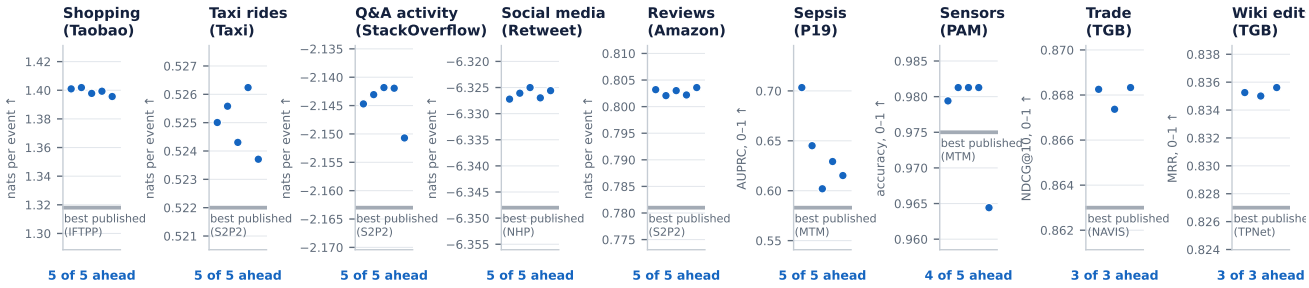
Benchmark	What the data is	Why it matters	Score (unit, direction)
Taobao	Clicks and purchases of users on Alibaba's Taobao marketplace, by product category	E-commerce: anticipating what a shopper does next and when	log-likelihood, nats per event ↑
Taxi	New York taxi pick-ups and drop-offs across city districts	Mobility and logistics: demand and fleet planning	log-likelihood, nats per event ↑
StackOverflow	Badges earned by users of the StackOverflow Q&A site	User engagement and lifecycle prediction	log-likelihood, nats per event ↑
Retweet	Retweet cascades on Twitter, by type of user retweeting	Social media: how information spreads, and how fast	log-likelihood, nats per event ↑
Amazon	Product reviews by Amazon customers, by category	Retail behaviour across product lines	log-likelihood, nats per event ↑
P19 sepsis	PhysioNet 2019: 38,803 intensive-care stays, 34 vital signs and lab values measured at irregular times	Early warning of sepsis, a leading cause of hospital deaths; every hour of earlier treatment matters	AUPRC and AUROC, 0 to 1 ↑
PAM	PAMAP2 physical-activity monitoring: 5,333 recordings from wearable inertial sensors (17 channels), 8 activities	Wearables and health monitoring: recognising what a person is doing from body-worn sensors	accuracy and F1, 0 to 1 ↑
P12 mortality	PhysioNet 2012: 11,988 intensive-care stays, 36 irregular measurements	Predicting in-hospital mortality from the first 48 hours	AUROC and AUPRC, 0 to 1 ↑
TGB trade	UN trade flows between 255 countries, 1986–2016	Which partners a country trades with most next year	NDCG@10, 0 to 1 ↑
TGB wiki	157,474 timed Wikipedia edits by 8,227 users on 1,000 pages	Which page a user edits next	MRR, 0 to 1 ↑
Neural-TPP benchmark (TMLR 2023/2025)	Seven event datasets (LastFM listening, MOOC course activity, Github, Stack Overflow, Wikipedia edits, MIMIC2 hospital visits, Retweets), five fixed splits each	A second, independent event benchmark: when the next event happens and what it is	negative log-likelihood, nats per sequence ↓

Reading the scores.

- **Log-likelihood (nats per event), higher is better.** It measures how much probability the model gave to what actually happened, both its timing and its type. A gain of 0.08 nats per event means the model assigns about 8% more probability to every real event, on average, compounding over every event in a sequence. Values can be negative; less negative is better.
- **AUPRC, higher is better (0 to 1).** How well the model ranks the rare patients who develop sepsis (4% of stays) ahead of everyone else, with emphasis on raising true alarms without false ones. It is the clinically relevant metric when the condition is rare. **AUROC** (0 to 1, higher is better) measures overall ranking.
- **Negative log-likelihood per sequence, lower is better.** The same probability measure summed over a whole sequence and negated, as the neural-TPP benchmark reports it; time and type ("mark") parts are reported separately and add up to the total.
- **NDCG@10 and MRR, higher is better (0 to 1).** How well the model ranks the true next partners or pages near the top of a list of all candidates.
- **Size and cost, lower is better.** Parameters measure how big a model is; multiply-adds per event measure the work needed for each prediction, which drives running cost and energy.

THE WINS

Ahead of the best published models



Each blue dot is one independent run (EasyTPP: five training seeds; sepsis and wearables: the five official data splits; temporal graphs: three pre-registered seeds), scored once on the sealed test set. The grey line is the best published result. Higher is better in every panel: log-likelihood in nats per event for the EasyTPP datasets, AUPRC (0 to 1) for sepsis, accuracy (0 to 1) for wearables, NDCG@10 and MRR (0 to 1) for the two temporal-graph benchmarks.

Benchmark	Best published	Sleeping Machines	Margin	Our size and compute vs the leading model
Taobao nats/event ↑	1.318 (IFTPP)	**1.399 ± 0.003**	+0.081	0.92× S2P2's compute per event; best time and type predictions
Taxi nats/event ↑	0.522 (S2P2)	**0.525 ± 0.001**	+0.003	**1/12 of S2P2's size and compute**; reproduced independently on separate hardware
StackOverflow nats/event ↑	-2.163 (S2P2)	-2.144 ± 0.004	+0.019	1.26× compute; a model of S2P2's size also wins (-2.153)
Retweet nats/event ↑	-6.348 (NHP)	-6.326 ± 0.001	+0.022	**1/15 of S2P2's size and compute**; S2P2 scores -6.365
Amazon nats/event ↑	0.781 (S2P2)	**0.803 ± 0.001**	+0.022	**0.29× S2P2's compute**, 3.6× smaller; best time and type predictions
Wearables, PAM accuracy ↑	0.975 (MTM)	**0.978 ± 0.007**	+0.003	**19× smaller** than MTM (46,316 vs 873K parameters); F1 0.980 vs 0.976; 4 of 5 splits ahead
Sepsis, P19 AUPRC ↑	0.583 (MTM)	**0.639 ± 0.039**	+0.056	62,681 parameters; AUROC 0.916 vs 0.903 ↑; without its temporal memory the same network drops to 0.572 on every split
Trade graphs, TGB tgbn-trade NDCG@10 ↑	0.863 (NAVIS)	**0.868 ± 0.0005**	+0.005	2,107 parameters (NAVIS 1,280); 3 of 3 sealed seeds ahead; a new domain
Wikipedia edits, TGB tgbl-wiki MRR ↑	0.827 (TPNet)	**0.835 ± 0.0003**	+0.008	7,995 parameters; 3 of 3 sealed seeds ahead; 4.5 ms per query on one CPU thread

± is the standard deviation across the runs (five; three on TGB). Compute is multiply-adds per event, counted from S2P2's released code at its published configuration.

One model for the whole benchmark. A single configuration of our model, with no per-dataset tuning, is ahead of the best published result on all five EasyTPP datasets: Taobao 1.397 vs 1.318, Taxi 0.526 vs 0.522, StackOverflow -2.144 vs -2.163, Retweet -6.324 vs -6.348, Amazon 0.802 vs 0.781 (nats per event ↑; 24 of 25 runs ahead; pre-registered protocol), at 0.07× to 1.03× the leading model's compute. Across all nine benchmarks, 40 of 41 independent runs are ahead.

A SECOND EVENT BENCHMARK

The same model, unchanged: the strongest type predictions, no total win yet

To test transfer, we took the frozen one-configuration EasyTPP model to the neural temporal-point-process benchmark of Bosser & Ben Taieb (TMLR 2023, 2025): seven datasets, five fixed splits each, every split scored once on its sealed test set. No setting was tuned for these datasets. The pre-registered bar is the best published time component plus the best published type component, a combination that no single published model reaches.

Dataset nats/sequence ↓	Bar	Sleeping Machines	Time / type parts (ours vs best published)	Verdict
Wikipedia edits	-122.62	$+11{,}421.66$	$+11{,}394.01$ vs -267.41 / **27.65 vs 144.79**	Behind: splits 1–4 average -277.54, but the guarded rerun of split 0 assigns exploding hazards to two test sequences; type part far ahead
MOOC course activity	-239.7	-226.85 ± 2.83	-298.57 vs -310.6 / 71.73 vs 70.9	Behind; timing gap
Stack Overflow	11.9	12.714 ± 0.874	-91.60 **vs** -91.1 / 104.31 vs 103.0	Behind; time part ahead
Github	-272.90	-270.62 ± 48.69	-390.93 **vs** -382.4 / 120.30 vs 109.5	Behind by 2.3; time part ahead; below every single published model's total
MIMIC2 hospital visits	2.42	7.01 ± 0.25	2.99 vs 0.13 / 4.02 vs 2.29	Behind
Retweets	-538.70	-515.16 ± 3.24	-598.81 vs -621.33 / 83.65 vs 82.63	Behind; type part level, timing gap
LastFM listening	-849.65	-774.64 ± 48.38	-1380.05 **vs** -1363.78 / 605.41 vs 514.13	Behind; time part ahead; below every single published model's total

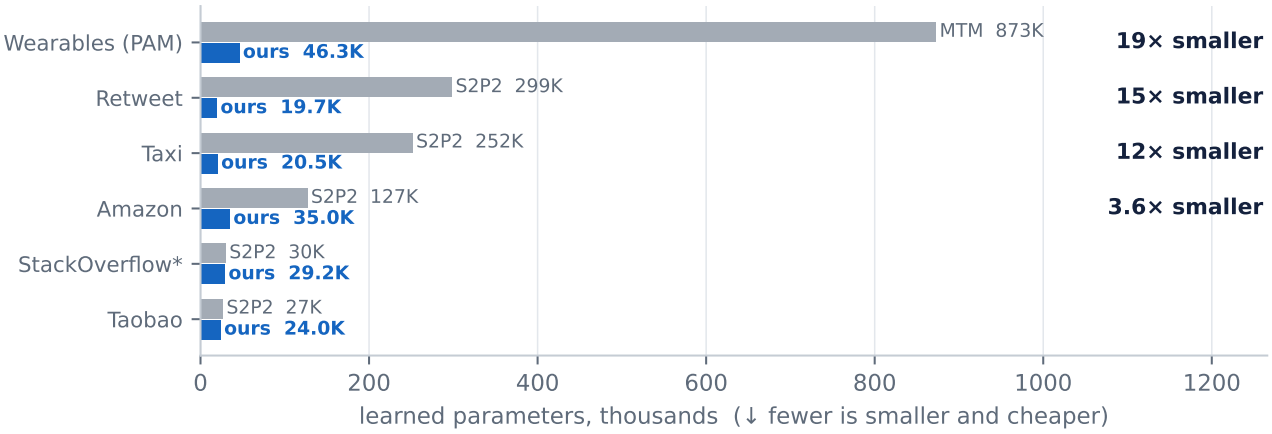
$\pm$ is the standard error over the five fixed splits. Negative log-likelihood per sequence: lower is better.

What Wikipedia shows. About 90% of consecutive edits return to the page just edited, and 10–27% of test pages never appear in training. Our addressed memory keeps one slot per page, written when that page occurs and read by that page's own score, so it can copy any page seen earlier in the sequence: type NLL 27.65 against 144.79 for the best published model, whose GRU scores pages as if guessing uniformly. The total is lost: one split's training failed numerically, and its pre-registered guarded rerun gives two test sequences hazards so confident that ordinary silences cost millions of nats. Bounding hazards is the development fix.

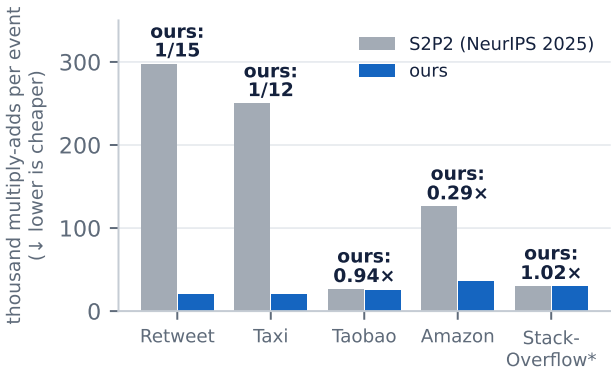
What the losses teach. A simple per-sequence counter, with no training, predicts the next type better than our model on two of these datasets, so the next development step adds persistent per-sequence type statistics to the memory. Github's training produced invalid values in four of five runs (one bad update poisoned the optimiser); those splits, and Wikipedia split 0, are retrained once with a guard that skips such updates, configuration unchanged, and scored once on test. The type-statistics step is development outside the sealed protocol.

EFFICIENCY

Winning while small



Learned parameters, in thousands; fewer means a smaller, cheaper model (lower is better). Grey: S2P2, the leading published state-space model on EasyTPP, at its published configuration for each dataset; for PAM, MTM (2025) at the parameter count it reports for PAM. Every Sleeping Machines model shown is also more accurate than the model beside it. *StackOverflow: the model matched to S2P2's size, which also wins.



Work per prediction, in thousands of multiply-adds per event (lower is cheaper). "ours: 1/15" means our model needs one fifteenth of S2P2's work.

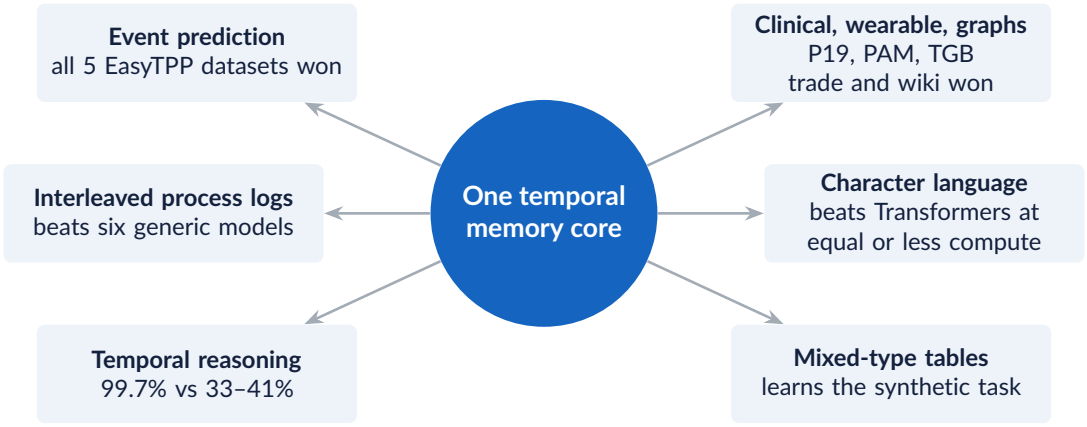
What size buys.

- **Running cost and energy** scale with the work per prediction.
- **Deployment anywhere**: models of 20,000 to 60,000 parameters run on ordinary CPUs, phones and embedded devices, close to the data.
- **Privacy**: clinical and industrial data can stay on site.
- **Headroom**: accuracy at small size means the architecture, not brute scale, is doing the work.

GENERALITY

One family across different kinds of data

S2P2 is shown on one kind of task. Our family wins on four kinds of data (event streams, clinical records, wearable sensors, interaction graphs) with shared machinery: the sepsis classifier reuses *the same temporal memory layer, same code and same size*, as the event-prediction model.



One shared core across four domains (development data, first seed). We trained a *single* temporal memory of 14,080 parameters on taxi rides, online shopping, Q&A activity and product reviews at the same time; only the small input and output layers differ per dataset. It matched or beat the four separately trained models on three of the four (log-likelihood, nats per event, higher is better: Taxi 0.490 vs 0.487, Taobao 1.290 vs 1.284, StackOverflow -2.186 vs -2.177), with about 40% fewer parameters in total. Mobility, shopping and online-community timing share one learned temporal dynamics. Amazon trailed and was still improving when training stopped; per-dataset training budgets are the next test.

Why generality matters for value. A company with one specialised model has one product. A company with a core that transfers across data types can serve many markets from one technology, and every new domain makes the core stronger.

WHY THE LARGER VISION IS ACHIEVABLE

Each layer of evidence supports the next

Next: scale and transfer. Large-scale language on GPUs, skill transfer between data types, clockless hardware.

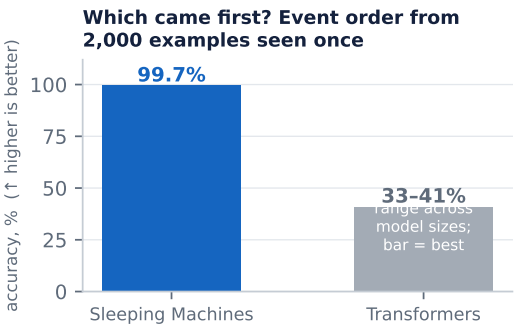
5. It generalises. One shared core across four event domains; one memory layer for events and clinical records.

4. It wins public benchmarks. All 5 EasyTPP datasets, P19 sepsis, PAM wearables and two TGB graph leaderboards, at up to 15× smaller size; 40 of 41 runs ahead.

3. It learns language efficiently. On 10M characters it beats a Transformer: 1.888 vs 1.908 bits per character ↓ with 0.40× its training compute. At 90M characters it beats a tuned Transformer-256×4 (1.783 vs 1.811 ↓) at 1.06× its compute.

2. The mechanisms work. Event order learned from 2,000 examples seen once: 99.7% accuracy vs 33–41% for Transformers ↑. Timing-only discrimination: 95.3% where any order-only model is capped at 50% ↑. Learning from roads not taken improves language 2.507 → 2.371 bits per character ↓ for 0.3% extra training work; more idle capacity improves it further at the same active work.

1. Mathematics: it contains today's leading architectures. A race of clocks reproduces the attention of Transformers *exactly*, and Mamba-style selective state-space models are exact members of the family (verified numerically to 4×10^{-13}). Everything those models can represent, ours can, plus native time and sparsity.



Why the steps add up. The family is not a narrow trick for one benchmark. Its mathematics contains the architectures that power today's AI; its mechanisms solve temporal problems those architectures fail at; it already learns language competitively at small scale; and it wins public benchmarks on four kinds of data with one shared memory design. Scale is the remaining work, and each layer beneath it is already demonstrated.

Hardware. In our cost model, the native language model moves 5.8× fewer bytes per character than a comparable Transformer at better quality (modelled, not yet measured on silicon; a funded milestone).

RIGOR

How we know the results are real

Sealed tests, rules fixed in advance. Every win uses the benchmark's official data splits. The reporting rule is written down before the test set is touched, and each model is scored on the test set exactly once.

The benchmark's own scorer agrees. Re-scoring our models with EasyTPP's own estimation method gives the same results within its sampling noise.

Independent runs. Benchmark headline results are averages of independent runs (five seeds or official splits; three pre-registered seeds on TGB), with its spread: 40 of 41 runs are ahead of the best published result.

We found and closed a scoring pitfall. Timestamps in these datasets are rounded (for example to whole seconds). Flexible models can exploit the rounding for large, fake gains. We built an audit that removes the rounding; we withdrew our own early results that failed it, and every reported win passes it.

Reproduced on separate hardware. The Taxi win was reproduced independently on a separate desktop computer with separately downloaded data: 0.5252 vs 0.5250 nats per event ↑, every run above S2P2. One-command kits let third parties rerun the EasyTPP and both TGB wins.

Losses reported as plainly as wins. Where we are behind, we say so with the numbers and the next test (next page). Comparisons always use the same units and protocol as the published result.

A NEW LEARNING RESULT

Forward computation teaches more economical credit

Ordinary learning uses error signals to train prediction. Our reciprocal design also uses information produced by prediction to improve credit assignment: which routes and learning updates will help on subsequent observations. Credit includes the evolving learner state, because an intervention can change future learning even after its direct effect has disappeared.

98.4%
lower counterfactual-credit error on eight fresh language prefixes

2.09×
faster complete corrected-credit acquisition on the measured pipeline

77.8%
less conditional gradient noise in a connected route controller

What was measured (11 October).

- **Fresh-prefix accuracy.** On eight new 48-target language prefixes, 65 candidate routes, diagonal optimizer-state transport gives contrast mean-squared error 1.927×10^{-10} versus 1.172×10^{-8} for static credit. All eight improve; the paired-prefix 95% improvement interval is positive. Error is 97.4% lower than reference-only adaptive credit.
- **Complete acquisition work.** Four prefixes, three rotated timing repeats: exhaustive independent learners take 26.933 seconds versus 12.893 seconds for transport plus exact factual correction. Both pay context and candidate discovery. Supported work falls from 165.164 to 10.775 GFLOPs (93.5% fewer); unsupported scalar/optimizer arithmetic remains included in wall time.
- **Connected-controller noise.** For a fixed 37,377-parameter controller on four earlier prefixes, conditional route-sampling gradient variance falls 77.8% versus static credit. Mean-gradient agreement is within 1.1×10^{-19}.

Status: a working research prototype. These are component results from a fixed pretrained two-layer native model, proper GPT-2 tokens and 64-dimensional adaptive lexical learners. A separate eight-prefix block-geometry test gives 80.3% lower credit error than diagonal transport, with 27.1% more kernel work. Online update-function learning beats its frozen twin by 0.00866 nats/token on eight fresh episodes (paired 95% interval positive, one initialization). Ordinary lexical Adam leads. Whole-native future-trained optimizer learning then improves DEV quality in all nine seed/depth cases (three seeds, depths2/4/8). It retains exact actor backprop and costs more CPU time; acquired-policy reuse is the next complete-work gate.

If this scales. More economical online adaptation could support models that track changing language, equipment, markets and user behaviour; local industrial learning; and sparse asynchronous hardware. The commercial test is better held-out prediction at lower complete learning cost. Deployment, whole-model speed, measured energy and economic impact require their own evidence.

How depth could benefit. Train connected credit functions jointly across layers, with shared forward information and longer-range consequence anchors. This could reduce compounding transport error; it must preserve useful credit gains rather than simply shrink signals. Measure prediction, error amplification, correction frequency and total work together as depth increases.

Source-bound completed results: `investment/forward_credit_evidence_20261011.json`. Transport metrics: one initialization, private TRAIN-tail observations. Native optimizer: three seeds on Taxi TRAIN/DEV, no TEST. A 64-dimensional state is not a 64-bit representation.

THE FRONTIER

Not ahead yet, or pending, and the next test

Front	Current result	Next test
P12 mortality	AUROC 0.871 vs 0.880 ↑ (behind); AUPRC level, 0.585 vs 0.586 ↑	Ensemble of five runs, still far smaller than the leader (in progress)
Tokenised language (GPT-2 tokens, FineWeb)	Ahead of the classic count model: 6.009 vs 6.537 nats per token ↓ at 1M training tokens (3 runs); at 4M tokens 5.510 vs 6.100, so the lead grows with data; recall across irregular gaps solved (97.5%)	Owned-model scaling and published-reference comparison under matching tokenizer, data and scoring
Large-scale language	At 90M characters a tuned Transformer-192×4 leads: 1.704 vs our 1.783 bits per character ↓	Integrated token-native temporal learning on CPU
Small-scale language	Tuned small LSTMs lead by 0.04–0.06 bits per character ↓ at 10M characters	Same programme as above
Real-world tables	Decision trees lead on the banknote dataset	Learned comparison discovery on real tables
FAS v2 (our anonymous-log benchmark)	Level on validation with a time-encoded Transformer reference: 0.702 vs 0.704 AUROC ↑ at about 1/7 of its parameters	Sealed test once the reference seeds finish; a tie is expected under the 0.02 rule
Second TPP benchmark (nats per sequence ↓)	All seven verdicts behind (page 5); type predictions far ahead on Wikipedia, time ahead on Stack Overflow, Github and LastFM	Remaining verdicts and guarded reruns; a developed attempt with per-sequence type statistics

WHAT COMES NEXT

From proof to platform

- **Publish** the complete event-leaderboard results, after the IP filing decision.
- **Scale language and learning**: integrated properly tokenized models on CPUs, compared with matching published references; test connected credit across depth.
- **Transfer**: show that skill learned on one kind of data helps on another.
- **First commercial vertical**: event-native models for interleaved, timestamped logs (industry, IoT, IT operations, security, finance) on existing CPUs and GPUs, the shortest path to revenue.
- **Hardware**: feasibility on FPGA and neuromorphic silicon for clockless, learning-capable event processors.

The investment case. A new model family, designed from first principles around time and sparse events, is already ahead of the best published models on nine public benchmarks across four kinds of data, including every dataset of the field's standard event benchmark, with models up to 15 times smaller; on a second event benchmark the same frozen model, untuned, predicts Wikipedia page types far better than any published model, while its totals still trail. Its mathematics contains today's dominant architectures. The remaining work is scale, transfer and hardware, and each has a concrete, measurable next step.